

Example 1. Let a be relatively prime to 6. Prove that $a^2 \equiv 1 \pmod{6}$.

Solution. We use the remainder r upon $a \div 6$ because:

Further we have proven

$$\gcd(a, 6) =$$

Thus, in order that $a \perp 6$, the possible remainders from $[0 \dots 5]$ are

$$a \equiv r \equiv$$

$$a \equiv r \equiv$$

□

Example 2. Let n be a positive integer and $d \mid n$.

Let a and b be integers such that $a \equiv b \pmod{n}$. Prove that $a \equiv b \pmod{d}$.

Solution. Because $a \equiv b \pmod{n}$ we have

$$a - b =$$

Because $d \mid n$ we have

$$n =$$

Thus

$$a - b =$$

Hence $a \equiv b \pmod{d}$.

□

Example 3. Let n and k be positive integers. Let a and b be integers. Prove that

$a \equiv b \pmod{n}$ if and only if $ka \equiv kb \pmod{kn}$

Solution. Observe that $a \equiv b \pmod{n}$ if and only if

which is equivalent, because k is nonzero, to

which occurs if and only if

□

A. **Division.** It is not always possible to divide without changing the modulus.

For example, we can divide all integers by 2, including the modulus to see

true congruence $4 \equiv 10 \pmod{6}$ is equivalent to:

But we **cannot** always skip dividing the modulus.

For we can see that:

Let n be a positive integer and a be an integer.

A **multiplicative inverse modulo n** of a is an integer a^{-1} so

Theorem. a has a multiplicative inverse modulo n if and only if:

Proof. We attempt to solve for x in the congruence

This congruence having a solution is equivalent to the Diophantine equation

having integer solution pair:

From our study of Diophantine equations, this occurs if and only if:

□

So for, example, 5 has a multiplicative inverse modulo 12.

$5 \cdot \quad \equiv 1 \pmod{12}$ and hence $5^{-1} \equiv \quad \pmod{12}$

As a trickier example, let's find the inverse of 37 modulo 31.

First apply the Euclidean algorithm, and reverse it, to eventually write

$$1 = 11 \cdot 101 - 30 \cdot 37$$

Then modulo 101 we have: $37^{-1} \equiv$

When we have a multiplicative inverse, we can “divide” by a by

Let n be a positive integer and $a \perp n$.

If b and c are integers so: $ab \equiv ac \pmod{n}$, then:

B. **Zero Cancellation.** This is another property that will not always hold.

Observe that

$$4 \cdot 3 \equiv 0 \pmod{12}$$

As a result, given $ab \equiv 0 \pmod{12}$ we **cannot** necessarily conclude

Let p be a prime and a and b be integers such that $ab \equiv 0 \pmod{p}$. Then

Proof. Because $ab \equiv 0 \pmod{p}$ it follows that

Because p is prime, it then must be the case

or equivalently

□

C. **Digits Divisibility.** Integers are by default presented with **digits** in **base 10**.

12342 =

Let us establish divisibility results using these digits.

Let n be an integer with base 10 representation

$$n = (a_m \cdots a_1 a_0)$$

$2 \mid n$ if and only if:

$5 \mid n$ if and only if:

$9 \mid n$ if and only if:

$11 \mid n$ if and only if:

Proof. Using its base 10 digits we have

$$n =$$

Taken modulo 2 gives

$$n \equiv$$

Taken modulo 5 gives

$$n \equiv$$

Taken modulo 9 gives

$$n \equiv$$

Taken modulo 11 gives

$$n \equiv$$

Each divisibility condition comes from the reduced expression.

□